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Treatment Adherence in ADHD: A Systematic Review of Influencing Factors and Strategies for Improvement

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Mobina Karimi[1], Sogand Ghasemzadeh[1], and Faezeh Shabanali Fami[1]

Abstract

Objective: The goal of this review was to more clearly understand what encourages or discourages treatment adherence in individuals with ADHD across different points in life. Instead of simply listing barriers, we set out to identify true, evidence-based strategies that could help patients and clinicians maximize long-term dedication to care.

Method: Following PRISMA guidelines, we systematically reviewed 52 peer-reviewed articles between 2014 and 2024. The primary sources used were five electronic databases (Scopus, Science Direct, PubMed, and Taylor & Francis). The studies included heterogeneous age groups—children, adolescents, and adults—and had diverse methodologies ranging from randomized controlled trials to qualitative interviews.

Results: Adherence to treatment for ADHD remains a problem, especially among teenagers. We organized the influencing factors into five main areas: sociodemographic, psychological/clinical features, medication-related issues, beliefs and attitudes, and broader systemic or environmental challenges. Conversely, interventions like pharmacological strategies, digital interventions and mobile-based reminders, psychoeducational and behavioral approaches, multimodal and long-term interventions, and consent-based strategies were all linked to improved adherence.

Conclusion: Improving adherence in ADHD requires a multifaceted, personalized approach that considers clinical, psychological, and contextual factors. Interventions that integrate pharmacologic and psychosocial strategies, while addressing barriers such as stigma, system fragmentation, and cultural mismatch, are essential to achieving better treatment outcomes and quality of life for individuals with ADHD. (*J. of Att. Dis. 2026; 30(5) 590-614*)

Keywords

medication adherence, ADHD, medication persistence, intervention, compliance

Introduction

ADHD ranks among the most prevalent neurodevelopmental disorders in children, affecting about 8% of this (Ayano et al., 2023; Bitsko et al., 2022). ADHD is characterized by a consistent pattern of inattention, hyperactivity-impulsivity, or a mix of both, which can hinder daily functioning and development (American Psychiatric Association, 2022). Those with ADHD frequently face challenges in organization, concentration, impulse control, planning, and reflective thinking. Additional traits include defiance, irritability, social challenges, and aggression (Brown, 2008, 2017; V. A. Harpin, 2005).

For many years, ADHD was perceived solely as a childhood condition that would dissipate upon reaching adulthood (Barkley, 1997; Brown, 2008, 2017). Yet, symptoms, especially inattention, are estimated to persist into adolescence and adulthood for approximately two-thirds of affected children (Lara et al., 2009). As individuals grow older, ADHD symptoms may evolve. The disorder usually continues into adulthood and can increase the risk of other psychiatric issues, resulting in negative outcomes like academic setbacks, job challenges, relationship difficulties, and a heightened risk of criminal behavior (Sayal et al., 2018). Given the substantial long-term adverse effects of ADHD on various life domains, including educational success and career stability, prompt treatment is vital (Arnold et al., 2020; Faraone et al., 2015; V. Harpin et al., 2016).

Common pharmacological interventions for ADHD include stimulant medications (like methylphenidate and

[1]University of Tehran, Iran

Corresponding Author:
Sogand Ghasemzadeh, Department of Psychology and Education of exceptional children, Faculty of Psychology and Education, University of Tehran, Tehran, Iran.
Email: s.ghasemzadeh@ut.ac.ir

amphetamines) and non-stimulant options (such as atomoxetine, clonidine, guanfacine, and viloxazine; Cortese, 2020). Stimulant medications are regarded as the primary treatment for individuals with ADHD (National Institute for Health and Care Excellence, 2018; Van Vyve et al., 2024) and are crucial in treating adolescents (Posner et al., 2020). Both stimulant and non-stimulant drugs effectively alleviate core ADHD symptoms (Bellato et al., 2025; Boland et al., 2020; Cortese et al., 2018; Radonjić et al., 2023) and enhance the overall quality of life for those affected (Bellato et al., 2025; Coghill et al., 2017; Groom & Cortese, 2022; Tsujii et al., 2020). However, their efficacy and safety in preschool children are limited, with lower effectiveness and higher rates of adverse side effects compared to older age groups, as evidenced by the Preschool ADHD Treatment Study (PATS; Kollins et al., 2006; Young et al., 2021).

Furthermore, pharmacological treatment promotes better family dynamics, academic results, school behavior, life skills, and social engagement, and reduces risky behaviors and any comorbid issues (Johnson et al., 2024). Beyond alleviating core symptoms, these medications also improve executive functions like planning, organization, working memory, and impulse control, facilitating better management of academic and job challenges (Groom & Cortese, 2022). Studies show reductions in behavioral issues, emotional stabilization, and decreased emotional dysregulation following medication use, which enhances emotional health, raises self-esteem, builds confidence, and fosters a positive self-image, ultimately improving quality of life (Adamo et al., 2015).

Psychosocial interventions also show promise and effectiveness in reducing ADHD's main symptoms (Tourjman et al., 2022). Such treatments encompass behavioral therapy, cognitive-behavioral therapy (CBT), parent training, neurocognitive training, social skills training, sleep enhancement, increased physical activity, and neurofeedback (Nazarova et al., 2022). While these interventions can be effective for managing ADHD symptoms in some cases, they are often recommended as adjuncts to stimulant medications to optimize outcomes, particularly for core symptoms, and to enhance secondary outcomes such as parenting strategies, social capabilities, academic success, and reductions in disruptive behaviors (Ogundele & Ayyash, 2023).

ADHD treatment usually involves a combination of therapies, including pharmacological and behavioral interventions (Ogundele & Ayyash, 2023). However, the Multimodal Treatment of ADHD (MTA) study demonstrated that a substantial number of individuals, particularly those with inattentive-type ADHD, can achieve positive outcomes with well-implemented non-pharmacological treatments alone, without requiring medications (Li, 2024; Qin, 2021). It is important to consider, adherence to these treatments varies greatly based on the type of intervention, personal traits, and environmental influences (Stewart et al., 2023).

Despite this, stimulant medications are often considered the gold standard for ADHD due to their efficacy in managing core symptoms, though issues like side effects, social stigma, and parental concerns about long-term use can lead to premature discontinuation. Conversely, non-pharmacological methods, such as cognitive-behavioral therapy (CBT), parent training, and neurofeedback, show effectiveness in enhancing emotional and behavioral outcomes, particularly for inattentive-type ADHD, though they require active participation from patients and caregivers, which can complicate long-term adherence (Charach & Fernandez, 2013; Kamimura-Nishimura et al., 2019).

Thus, it is crucial to compare adherence rates across various treatment types and understand the factors affecting these rates to grasp the effectiveness of interventions in real-world scenarios. Early diagnosis and consistent adherence to both psychological and pharmacological treatments are critical for the long-term management of ADHD. Nevertheless, poor adherence is a frequent challenge in chronic conditions like ADHD (Safavi et al., 2019). The World Health Organization (2003) defines treatment adherence as how well an individual's behaviors, such as taking medication, adhering to diets, and implementing lifestyle changes, align with healthcare provider recommendations. To boost treatment adherence promptly, evaluating and strengthening related factors before initiating treatment is vital. These factors, which should guide treatment decisions, encompass patient and parental attitudes toward treatment, trust in the doctor-patient relationship, family support, perceived medication benefits, stigma, understanding of the disorder and treatment options, and medication side effects (Ferrin et al., 2012; Gajria et al., 2014; Niemeyer et al., 2018; Wolraich et al., 2019).

For medication adherence, contradictory information about the drug, dosage frequency, fears regarding growth effects, addiction risks, doubts about effectiveness, and concerns about side effects are among the most significant factors influencing adherence (Kamimura-Nishimura et al., 2019; Treuer et al., 2016). Additional issues, such as medication side effects, lack of symptom management from medication, societal stigma surrounding medication use, forgetting to take medication, and dissatisfaction with the medication experience, are also correlated with lower adherence rates (Barnard-Brak et al., 2023; Gajria et al., 2014). Treatment adherence patterns may also differ by age group (children, adolescents, and adults). Children and adolescents, being more dependent on their parents, are often swayed by their parents' treatment attitudes. In contrast, adults may encounter alternative challenges, such as social stigma or adverse effects from treatments, especially pharmaceutical ones (Ayano et al., 2023; Campagna et al., 2020). However, there is a significant gap in research systematically exploring these age-related differences, leaving uncertainty about which factors most strongly impact treatment decisions at each developmental stage.

Recognizing these differences is essential for developing age-specific interventions that enhance adherence and improve long-term treatment outcomes.

Treatment adherence is vital for managing ADHD symptoms and minimizing the long-term negative effects associated with the disorder. Longitudinal studies suggest that individuals who adhere to treatment during childhood and adolescence achieve better academic, occupational, and social outcomes in adulthood. Conversely, poor adherence correlates with higher risk-taking behaviors, lower educational achievements, and challenges in emotional regulation. Therefore, early identification and intervention strategies to improve adherence can provide significant benefits for individuals with ADHD (Abdallah et al., 2024; Shaw et al., 2012).

Various interventions have been proposed to improve treatment adherence, including educational programs for individuals with ADHD and their families, digital technologies such as medication reminder apps, and regular counseling sessions with healthcare professionals (Kamimura-Nishimura et al., 2019; Parkin et al., 2022). However, the evidence regarding their effectiveness remains scattered, necessitating a more comprehensive review. Existing research often examines pharmacological and non-pharmacological treatments separately, limiting a holistic understanding of the factors influencing adherence across different age groups. This fragmented perspective hinders the development of targeted interventions tailored to the specific needs of individuals with ADHD. Although studies have explored the reasons for treatment discontinuation, a systematic review synthesizing these findings can provide valuable insights for researchers, clinicians, and policymakers.

Therefore, this study aims to address this gap by examining the factors that influence treatment adherence in individuals with ADHD, ultimately contributing to improved treatment management and better long-term outcomes. Specifically, this study seeks to achieve three objectives:

1. To systematically examine the factors influencing treatment adherence (both medication and psychosocial interventions) in ADHD patients across different age groups (children, adolescents, and adults).
2. To review and evaluate the effectiveness of existing adherence-enhancing strategies for ADHD treatment, including technological and behavioral interventions.
3. To compare adherence patterns and their determining factors between pediatric (<18 years) and adult (≥ 18 years) ADHD populations.

Method

Study Design

This systematic review was conducted following the updated reporting guidelines for systematic reviews and meta-analyses (PRISMA 2020) to ensure transparency and methodological rigor (Page et al., 2021). A comprehensive literature search was performed across four major electronic databases: PubMed, ScienceDirect, Scopus, and Taylor & Francis Online, covering the period from January 1, 2014, to December 31, 2024. The search strategy included a combination of keywords related to ADHD, treatment adherence, and interventions to improve adherence. Specifically, the following search terms were used: "ADHD or attention-hyperactivity deficient," "medication or pharmacotherapy or treatment or psychological or psychopharmacological," and "adherence or compliance or persistent or non-discontinuation barrier." The search was limited to studies published in English to maintain consistency in data interpretation.

Screening and Selection Process

The screening process was conducted in multiple stages. Initially, duplicate records were removed. Subsequently, two independent researchers applied the inclusion and exclusion criteria at the title and abstract levels to filter relevant studies. Full texts of eligible studies were then retrieved and reviewed based on predefined selection criteria. If a study was excluded at this stage, the reason for exclusion was documented. To ensure a comprehensive inclusion of relevant studies, primary studies identified in previous systematic reviews were manually reviewed to check for any additional studies not captured in the initial database search. The entire screening process is illustrated in Figure 1 (PRISMA flowchart). One researcher conducted the screening, and any discrepancies regarding study inclusion were resolved through discussions with other research team members.

Inclusion and Exclusion Criteria

Studies were included if they met the following criteria:

1. Diagnosis of ADHD based on any edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM) or as determined by a psychiatrist or child psychologist. There were no restrictions on ADHD subtypes, gender, intelligence level, or socioeconomic status.
2. Implementation of non-pharmacological interventions or the prescription of stimulant, non-stimulant, or other medications for ADHD treatment, regardless of treatment duration.
3. Reporting at least one outcome related to treatment adherence for any intervention.
4. Inclusion of children, adolescents, adults, or parents/caregivers as study participants.
5. Reporting on treatment adherence outcomes or reasons for discontinuation, switching, and/or adherence to ADHD pharmacotherapy.

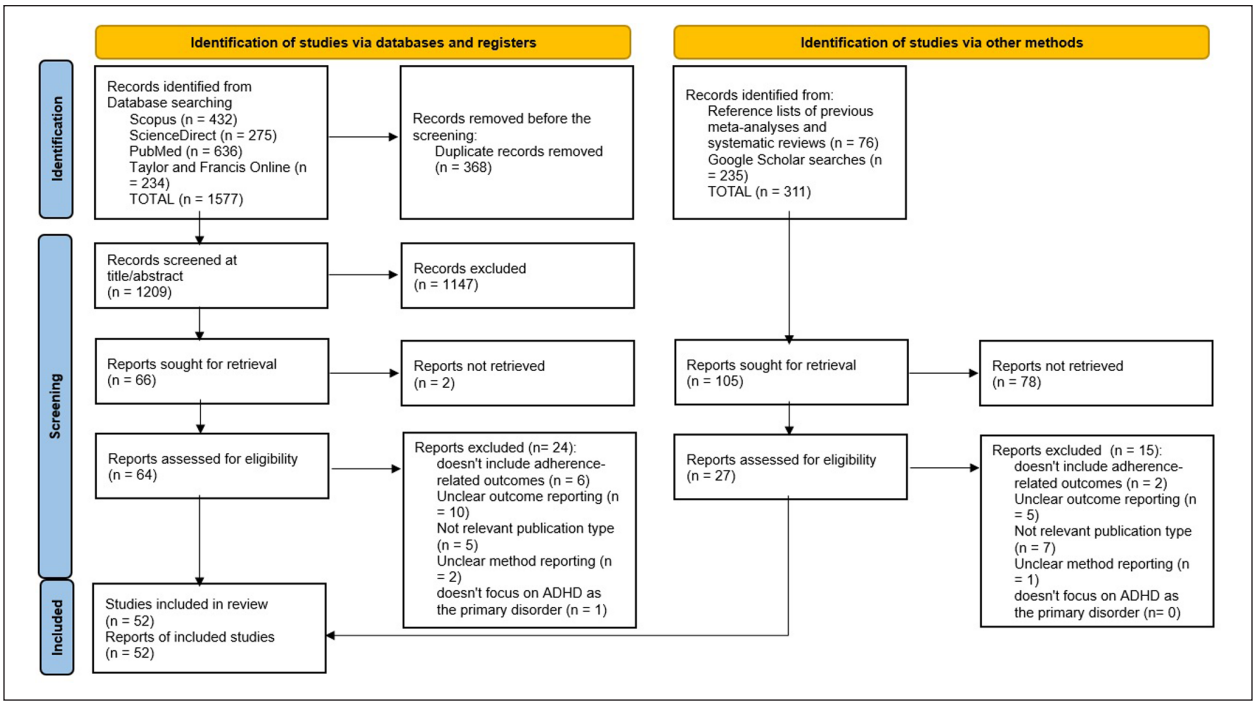


Figure 1. Study selection flow diagram.

Studies were excluded if they met any of the following criteria:

1. Not published in English.
2. Lacked clear data or results.
3. Classified as conference papers, books, book chapters, addresses, bibliographies, biographies/lectures, case reports, author opinions, letters, systematic reviews/meta-analyses, theses, or dissertations.
4. Focused on disorders other than ADHD.
5. Involved participants who did not receive pharmacological treatment.

Data Extraction

Data extraction was conducted independently by two researchers using a standardized data extraction form to ensure consistency and accuracy. The extracted data included:

- Study characteristics: author names, year of publication, country, and study type.
- Participant details: age, population type, and sample size.
- Intervention details: type of intervention, components, and outcomes (if applicable).
- Factors influencing treatment adherence.
- Measurement tools and adherence assessment methods.

In case of discrepancies between the researchers, disagreements were resolved through discussion, and if necessary, a third or fourth researcher was consulted to reach a consensus. The extracted data were summarized in tables and analyzed for interpretation.

Data Synthesis and Analysis

The primary outcomes assessed in this review were: Factors influencing ADHD treatment adherence, and the Impact of interventions on improving adherence to ADHD medication. A qualitative synthesis of existing studies was conducted to identify and analyze various individual, environmental, and treatment-related factors affecting treatment adherence. These factors were examined based on whether they acted as facilitators or barriers to adherence. Additionally, a qualitative comparative analysis was performed to explore differences in adherence patterns between children and adolescents (under 18 years) versus adults (18 years and older). This analysis identified distinct trends in adherence, influencing factors, and reasons for treatment discontinuation across age groups. Common reasons for medication discontinuation—such as side effects, neglect of treatment, and medication switching—were systematically categorized based on their reported frequency in the included studies. Finally, this review examined various strategies implemented in previous studies to enhance ADHD treatment adherence, including behavioral, educational, psychological, and technology-based interventions.

These strategies were evaluated based on their reported effectiveness and summarized in the review findings.

Results

A total of 52 articles were included in this systematic review. A summary of these studies, including types of studies, sample size, participant age, and other relevant characteristics, is presented in Table 1. About 35% of the studies in this review consisted of longitudinal and retrospective cohort studies or chart reviews. Another 21% were cross-sectional studies, and another 21% were experimental studies of different sorts, including randomized control trials and non-randomized intervention studies. Then, about 13% were qualitative investigations, while less than 1% were survey-based studies. Two studies, again, less than 1%, employed the mixed-method approach. Sample sizes ranged widely, from as small as 10 to over 3 million participants, averaging just over 66,000 and a median value of 145, totaling close to 3.38 million throughout all studies. Participant ages ranged widely from 5 to 65 years, with a median age of about 14.5 and an interquartile range of 10.75 years. A noteworthy point to note is that the definitions and measurements of dependent and independent variables varied greatly across the studies, as will be further detailed in Table 1.

After classifying the results, this systematic review identified 40 factors (group A) and 12 strategies (group B) influencing ADHD treatment outcomes among the chosen articles. Influential factors (group A) included five groups: demographic and socioeconomic (gender, age, race/ethnicity, socioeconomic status, educational level, and family structure), clinical and psychological conditions (comorbidities, symptom severity, symptom type, family history, age at diagnosis, and mode of delivery), medication-related factors (medication type, formulation, switching, dosage and adjustments, side effects and benefits, ineffectiveness, and co-medication and concomitant), and psychological factors and beliefs (parental and patients' beliefs, perception of ADHD, perceived purpose of treatment, sense of self and identity, necessity–concern balance, and personality traits), and external and systemic factors (family dynamics, social support and stigma, lifestyle and practical barriers, healthcare system and providers, financial and insurance barriers, and medication knowledge). These factors shaped treatment decisions, adherence, and long-term outcomes. Table 2 summarizes the ranked categories of factors affecting ADHD treatment adherence, integrating both the frequency with which each category appeared in the literature and its relative importance in predicting treatment persistence.

In addition, key interventions (group B) can be classified into five main categories: pharmacological strategies, digital interventions and mobile-based reminders, psychoeducational and behavioral approaches, multimodal and long-term interventions, and consent-based strategies. The highlighted items mentioned above are presented below as separate headings, in the order in which they were introduced.

Factors Influencing Treatment Adherence in ADHD

Demographic and Socioeconomic Factors. Different studies found that demographic and socioeconomic characteristics significantly influenced ADHD medication adherence. Figure 2 summarizes the findings in this section.

Gender. Gender differences were described inconsistently, with some studies describing higher adherence in women (Ishøy et al., 2024; Mohr-Jensen et al., 2020; Rao et al., 2021; Sobanski et al., 2014), while others described higher adherence in men (Ishizuya et al., 2021; Kamimura-Nishimura et al., 2022; Lee & Je, 2018). These discrepancies suggest that gender adherence patterns may be determined by age, treatment setting, and socioeconomic status.

Age. Similarly, age was a significant factor. The impact of adherence and age was heterogeneous between studies; a general trend of decreasing adherence with increasing age has been observed (Ayaz et al., 2014; Lee & Je, 2018; Sleath et al., 2017; Tuncturk et al., 2023). Some research found higher adherence during childhood, since parents arranged drug regimens such that drugs were taken regularly (Brinkman, Sucharew, et al., 2018; Kamimura-Nishimura et al., 2022). A few studies indicated that when adolescents gain independence and medication responsibility shifts from their parents to them, adherence decreases, particularly at college or work transition periods (Brinkman, Sucharew, et al., 2018; Charach et al., 2014; Khan & Aslani, 2020; Wagoner et al., 2020). Only one study identified that treatment adherence is greater during late adolescence than during adolescence and late childhood (Bhang et al., 2017). Adult adherence findings were inconclusive. Some studies indicate that adherence continues to decline in adulthood, as adults tend to regard ADHD symptoms as less impairing or they form compensatory tactics, reducing reliance on medication (Ishøy et al., 2024; Jeun et al., 2024). However, other studies indicated that adherence stabilizes or even rises in adulthood, particularly among those individuals who find benefits of medication toward occupational and daily functioning (Khan & Aslani, 2021).

Race/ethnicity. Race and ethnicity also influenced adherence. Non-whites had significantly lower levels of adherence compared to white patients (Barnard-Brak et al., 2023; Kamimura-Nishimura et al., 2022; Rao et al., 2021; Sleath et al., 2017) and were at higher risk for premature discontinuation and two or more medication gaps (Ji et al., 2018). These findings indicate structural and cultural obstacles,

Table 1. Summary of Studies Main Characteristics.

Authors	Country	Study design	Patients age	Sample size	Independent variable[a]	Participants[b]	Adherence measures[c]	Additional measures[d]	Factors/ strategies
Jeun et al. (2024)	USA	Cross-Sectional Study	18–25	798	G1, G2, G3, G5	P2	T2	11	F
Baykal et al. (2024)	TR	Retrospective Design	6–12	132	G1, G2, G3, G4, G5	P1, P3	T4	19, 12, 18	F
Ishøy et al. (2024)	DK	Cohort Study	18–50+	23916	G1, G2, G3, G5	P2	T3	11, 14, 16	F
Alsubaie et al. (2024)	SA	Cross-Sectional Study	6–8	221	G1, G3, G4	P1, P3	T1	13, 11, 17, 15	F
Murat and Tekin (2023)	TR	Cross-Sectional Study	8–13	98	G1	P1	T1	11, 12, 19	F
Tuncturk et al. (2023)	TR	Retrospective Design	6–17	518	G1, G3	P1	T3	11	F
Barnard-Brak et al. (2023)	USA	Survey Design	13–24	196	G1, G3, G4, G5	P1, P2	T1	11	F
Carvalho et al. (2023)	BR	RCT	18–45	73	G6	P2	T2	13	S
Barnard-Brak amd Kudesey (2022)	USA	Survey Design	12–25	157	G3	P1, P2	T1	17	F
Titheradge et al. (2022)	UK	Semi-Structured Interview	14–29	64	G2, G3, G4, G5	P1, P2	T1	18	F
Schein et al. (2022)	USA	Retrospective Chart Review	6–17	434	G3, G4, G5	P1	T1	11, 14, 18	F
Kamimura-Nishimura et al. (2022)	USA	Multiple Randomized Trial Design	7–11	144	G1, G2, G3, G4, G5	P1	T2	11, 12, 17, 13	F
Rao et al. (2021)	USA	Cohort Study	17–19	4,466	G1, G5	P1	T2	11, 17	F
Nayak et al. (2021)	IN	Cross-Sectional Study	5– 12	50	G2, G3	P1	T1	12	F
Khan and Aslani (2021)	AU	Focus Group Interview	18–65	20	G3, G4, G5	P2	T1	18, 14	F
Ishizuya et al. (2021)	JPN	Cohort Study	6–17	1,353	G1, G3	P1	T3, T4	11	F
Cheung et al. (2021)	NL	Cohort Study	5–16	307	G1, G3	P1	T2, T3	11, 14	F
Fried et al. (2020)	USA	Non-Randomized Study	6–12	333	G6	P1	T2	11	S
Wagoner et al. (2020)	USA	Observational Study	17–21	45	G3, G4, G5	P2	T5	11, 14, 15	F
Mohr-Jensen et al. (2020)	DK	Cross-Sectional Study	18–60	3,165,844	G1, G2, G3, G5	P2	T3	14, 11, 16	F
Khan and Aslani (2020)	AU	Focus Group Interview	12–18	34	G3, G4, G5	P1, P3	T1	18, 14	F
Zheng et al. (2020)	USA	RCT	6–11	211	G6	P1, P3	T1	13, 12	S
Emilsson et al. (2020)	SE	Cross-Sectional Study	13–17	99	G3, G4	P1	T1	13	F
Boudreau and Mah (2020)	CA	Quantitative Prospective Design	6–13	69	G4	P1	T1	13	F
Safavi et al. (2019)	IR	Cross-Sectional Study	6–12	118	G1, G2, G4, G5	P1	T1	13, 12, 11	F
Biederman et al. (2019)	USA	Non-Randomized Study	18–55	552	G6	P2	T2	11, 12	S
Nagae et al. (2019)	JPN	Non-Randomized Study	8–15	41	G6	P1, P3	-	14, 12, 15	S
Brinkman, Simon, and Epstein (2018)	USA	Cohort Design	7–10	372	G3, G4, G5	P1	T1	14, 12, 11	F
Lee and Je (2018)	KOR	Cross-Sectional Study	6–18	3,317	G1, G2, G3	P1	T3	19	F
Ji et al. (2018)	USA	Cohort Study	6–12	102,669	G1, G2, G3	P1	T3	11	F
Druedahl and Kälvemark Sporrong (2018)	DK	Semi-Structured Individual and Focus Group Interviews	22–29	10	G3, G4, G5	P2	T1	18	F
Brinkman, Sucharew, et al. (2018)	USA	Cohort Study	6–10	89	G1, G2, G4, G5	P1	T3	16, 15, 12, 13	F
Weisman et al. (2018)	IL	RCT	6–16	39	G6	P1	T5	12, 19	S
Gould and Doucette (2018)	CA	Survey Design	18–22	53	G4, G5	P2	T1	11, 13	F
Sleath et al. (2017)	USA	Mixed-Method Design	7–17	70	G1, G5	P1, P3	T1	11, 13	F
Schaefer et al. (2017)	USA	Qualitative Interview	18–21	10	G3, G4, G5	P2	T1	18, 11, 14	F
Kosse et al. (2017)	NL	Cross-Sectional Study	12–18	181	G4, G5	P1	T1	13	F
Bhang et al. (2017)	KOR	Cohort Study	6–18	69,631	G1, G2, G3, G5	P1	T2	11	F
Emilsson et al. (2017)	SE	Cross-Sectional Study	13–17	101	G3, G4	P1	T1	13	F
Meyers et al. (2017)	USA	Retrospective Cohort Design	6 –17	165	G6	P1	T2	17	S
Enns et al. (2017)	CA	Cohort Study	5–17	2,369	G6	P1	T2	-	S

(Continued)

Table 1. (continued)

Authors	Country	Study design	Patients age	Sample size	Independent variable[a]	Participants[b]	Adherence measures[c]	Additional measures[d]	Factors/ strategies
Naenen-Hernani et al. (2017)	ES	Observational, Cohort Study	6–15	141	G6	P1	T3	I2, I10, I9	S
Semerci et al. (2016)	TR	Retrospective Chart Review Study	18–55	102	G1, G2, G3	P2	T4	I1, I9, I2,	F
Antony (2016)	IN	Cross-Sectional Study	6–12	40	G2, G3, G5	P1	T4	I4, I9	F
Hogue et al. (2016)	USA	Non-Randomized Study	13–17	35	G6	P1, P3	T2	I6	S
Bai et al. (2015)	CN	RCT	6–16	89	G6	P3	T1, T2	I3, I2	S
Montoya et al. (2014)	ES	RCT	6–12	270	G6	P1, P3	T5	I2, I9, I5, I4	S
Sobanski et al. (2014)	GER	Placebo-Controlled Trial	18+	241	G1, G2, G3	P2	T5	I2	F
O'Callaghan (2014)	USA	Mixed-Method Design	18–64	85	G3, G5	P2	T1	I5, I1, I3	F
Charach et al. (2014)	CA	Semi-Structured Interview	12–15	24	G4, G5	P1, P3	T1	I8	F
Ayaz et al. (2014)	TR	Retrospective Design	6–18	877	G1, G2, G3, G4, G5	P1	T3	I2, I9	F
Ahmed et al. (2017)	AU	Focus Group Interview	—	16	G3, G4, G5	P3	T1	I8	F

[a]G1: Demographic and Socioeconomic Factors; G2: Clinical and Psychological Conditions; G3: Medication-Related Factors; G4: Psychological Factors and Beliefs; G5: External and Systemic Factors; and G6: Treatment and Interventions

[b]P1: Children and Adolescents (0–18 years); P2: Young Adults and Adults (19–55 years); and P3: Parents, Caregivers, and Educators

[c]T1: Self-Reported Adherence Measures (e.g., Medication Adherence Report Scale [MARS], Morisky Medication Adherence Scale, and Compliance Rating Scale [CRS], Self-reports, parent-reports, and physician-reported adherence); T2: Prescription Refill and Possession-Based Adherence (e.g., Medication Possession Ratio [MPR], Proportion of Days Covered [PDC], and refill-based adherence with common cutoffs 80%); T3: Treatment continuation without gaps and Discontinuation; T4: Clinic-Based Adherence Measures (e.g., clinic visits, follow-up appointments, and progress reports); and T5: Pill Counts and Direct Medication Tracking

[d]I1: Sociodemographic and Clinical Characteristics (e.g., Tools like the Health Insurance Review and Assessment [HIRA] claims database and the Medical Expenditure Panel Survey [MEPS]); I2: ADHD Symptom and Clinical Rating Scales (e.g., ADHD Rating Scale [ADHD-RS], Vanderbilt ADHD Diagnostic Parents Rating Scale [VADPRS], and Conners' Parent Rating Scale); I3: Beliefs, Attitudes, and Perceptions about ADHD and Treatment (e.g., Beliefs about Medicines Questionnaire [BMQ] and Brief Illness Perceptions Questionnaire [BIPQ]); I4: Treatment, Medication, and Adherence (e.g., prescription registries, treatment satisfaction questionnaires, and adherence checklists); I5: Quality of Life, Functioning, and Patient Experience (e.g., Pediatric Quality of Life Inventory [PedsQL] and Weiss Functional Impairment Rating Scale [WFIRS-P]); I6: Psychiatric Comorbidities and Psychological Factors (e.g., Comprehensive Addiction Severity Index for Adolescents [CASI-A] and Mini International Neuropsychiatric Interview [MINI]); I7: Healthcare Utilization, Costs, and Side Effects (e.g., healthcare utilization and cost assessments, Emergency Department [ED] visits, and side effect monitoring); I8: Qualitative Interviews and Focus Groups; I9: Clinical Global Impression and Diagnostic Tools (e.g., Clinical Global Impression Scale [CGI] and DSM-IV-based screening); and I10: Statistical and Risk Measurement Tools (e.g., Relative Risk Reduction [RRR] and Number Needed to Treat [NNT])

Table 2. Ranking of Factors Influencing ADHD Treatment Adherence.

Rank	Group	Frequency	Relative importance
1	G3—Medication-related factors	31	Very high importance
2	G5—External and systemic factors	26	High importance
3	G4—Psychological factors and beliefs	22	Moderate to high importance
4	G1—Demographic and socioeconomic factors	21	Moderate importance
5	G2—Clinical and psychological conditions	16	Lowest importance

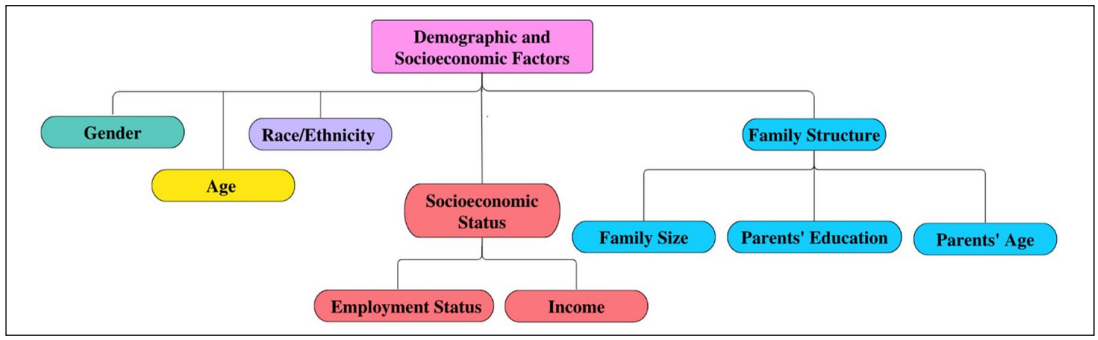


Figure 2. Subdimensions of the demographic and socioeconomic factors.

including healthcare access, racial discrimination, and differing expectations about the treatment of ADHD.

Socioeconomic Factors. Socioeconomic factors were strongly correlated with adherence outcomes. Higher educational level (Mohr-Jensen et al., 2020; Semerci et al., 2016; Sobanski et al., 2014) and employment status (Ishøy et al., 2024; Mohr-Jensen et al., 2020) were linked with higher medication continuity, while lower family income was linked with higher discontinuation rates (Kamimura-Nishimura et al., 2022; Rao et al., 2021; Safavi et al., 2019). Cheung et al. (2021) believe that median income is more positively associated with treatment adherence than very high income. Similarly, households of lower incomes also reported more adverse attitudes toward the use of ADHD medication (Alsubaie et al., 2024), and financial insecurity (i.e., incapacity benefits and income support) predicted a higher risk of discontinuation (Mohr-Jensen et al., 2020).

Family Structure. Reduced parental assessment of ADHD severity and increased negative medication beliefs also led to decreased medication coverage rates (Kamimura-Nishimura et al., 2022). Alsubaie et al. (2024) found that maternal age was demonstrated to influence beliefs about ADHD treatment and drug-taking adherence. Older mothers had more negative medication and treatment beliefs. These findings suggest that older mothers may be more skeptical about ADHD pharmacological treatment, which may lead to lower adherence in their children. Increased paternal and maternal education was associated with increased child adherence rates, too (Alsubaie et al., 2024; Murat & Tekin, 2023). Cheung et al. (2021) also found that the number of children in the household was associated with ADHD medication adherence. This would mean that the greater the number of children, the lower the parental ability to monitor drug schedules intensely, cope with side effects, and maintain adherence behaviors. Parents with more children may also have greater financial and care responsibilities, which may affect treatment consistency.

Compared with other groups, the effect of these factors was at a moderate level. Overall, demographic factors, although associated with noticeable variations in adherence in some studies, generally exerted a moderate or weaker influence compared to medication-related factors and the healthcare system. This category appears to act more as a background or moderating influence rather than a primary predictor of treatment continuation.

Clinical and Psychological Conditions. Six clinical and psychological predictors of adherence to ADHD medication have been identified in this systematic review. This category exerts less influence than others, its role was more pronounced in high-risk patients. Figure 3 presents a summary of the findings in this section.

Comorbid Conditions. Comorbid psychiatric illnesses contributed remarkably to treatment persistence (Baykal et al., 2024; Bhang et al., 2017; Ji et al., 2018; Titheradge et al., 2022). While there were some conditions, such as Oppositional Defiant Disorder (Kamimura-Nishimura et al., 2022),

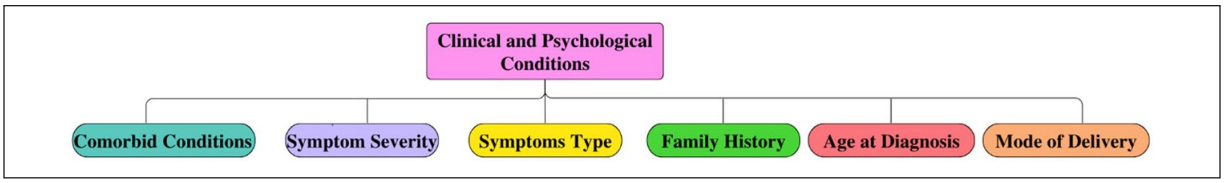


Figure 3. Subdimensions of the clinical and psychological conditions.

tic disorders, epilepsy, developmental delay, intellectual disabilities (Ishøy et al., 2024; Lee & Je, 2018)Eating disorders, conduct disorder, and sleep disorders (Ishoy et al.,2024), are associated with higher adherence, while others, such as depression (Lee & Je, 2018)Specific learning disorder (Baykal et al., 2024), Substance use disorders, schizophrenia spectrum disorders, and affective disorders were associated with higher discontinuation rates (Ishøy et al., 2024). Second, individuals with greater than one mental health comorbidity were less adherent to their ADHD medication compared to those with no comorbidities (Jeun et al., 2024). These findings suggest that certain comorbidities enhance the perceived demand for medicine and others pose additional barriers to adherence.

Symptom Severity. The Severity and nature of ADHD symptoms also determined patterns of adherence. Findings regarding the severity of symptoms and adherence varied across studies. Nayak et al. (2021) presented that longer duration of illness and more severe illness were both inversely related to adherence, indicating that patients with greater ADHD history or more severe symptoms were less likely to adhere to the drug regimen. Also, Sobanski et al. (2014) illustrated that Mild to moderate baseline ADHD symptoms were associated with non-adherence. In addition, symptom generalizability across functional domains and Illness Consequences was negatively related to adherence (Emilsson et al., 2017; Semerci et al., 2016) showing that patients with longer durations of ADHD, greater severity of symptoms, or more far-reaching functional impairment were less adherent to their medication regimen. Also, as measured by the CGI scale, one study showed that higher CGI severity scores were linked with higher adherence, and therefore, those with severe symptoms were more likely to consider medication essential to daily life (Baykal et al., 2024). But other studies indicated that higher CGI severity scores were linked with lower adherence (Ayaz et al., 2014; Semerci et al., 2016), possibly due to heightened side effects, lower satisfaction with the effectiveness of treatment, or greater frustration with taking medication.

Symptom Type. Patients with greater hyperactivity/impulsivity symptoms were more likely to be compliant with medication (Ayaz et al., 2014; Safavi et al., 2019), possibly because symptom control was clearer with pharmacological treatment. In contrast, greater inattention symptoms were linked to lesser adherence, perhaps because they forgot and found it hard to follow a consistent medication regimen.

Family History. Family history of ADHD and psychiatric disorders also had an impact on adherence. Subjects with a history of ADHD in the family or prior exposure to psychotropic treatment were more prone to adherence to medication, likely because they were more familiar and accepting of drug therapy. Conversely, subjects without a family history of ADHD were more likely to discontinue medication, possibly because of stigma or disbelief regarding ADHD treatment (Baykal et al., 2024; Kamimura-Nishimura et al., 2022; Safavi et al., 2019; Sobanski et al., 2014).

Age at Diagnosis. An earlier age at diagnosis is associated with higher rates of treatment continuation (Baykal et al., 2024), whereas earlier treatment initiation, although initially beneficial, can lead to higher chances of discontinuation, with late-diagnosed patients more likely to continue treatment (Mohr-Jensen et al., 2020).

Mode of Birth. Additionally, mode of birth was identified as a potential predictor, with one study stating that delivery via cesarean section could be linked with lower adherence levels, implying early-life factors could influence long-term treatment participation (Antony, 2016). Importantly, the term mode of birth here refers exclusively to childbirth methods and should not be confused with the pharmacological term route of administration, which describes different forms of medication delivery (e.g., oral, intravenous, and transdermal).

Medication-Related Factors. In this review, eight factors influencing medication adherence among patients with ADHD were uncovered. Figure 4 presents a summary of the findings in this section.

Medication Type. Among the most significant factors was the medication type; children who take stimulants (Nayak et al., 2021) like Methylphenidate Immediate Release (IR) and Methylphenidate Extended Release (ER), and non-stimulants like Atomoxetine and Aripiprazole/Imipramine, have been shown to have good adherence (Alsubaie et al., 2024). Atomoxetine has been more frequently reported to

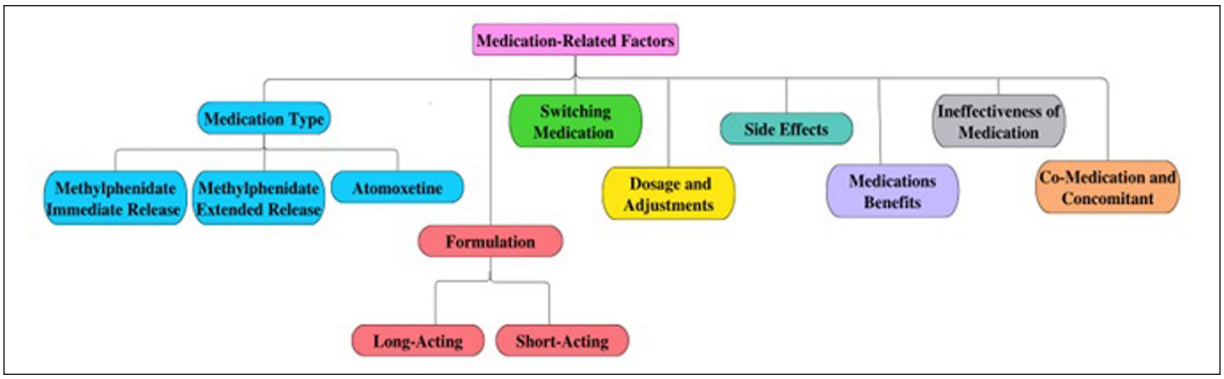


Figure 4. Subdimensions of the medication-related factors.

have higher adherence than short-acting methylphenidate, which could be due to its once-daily dosage and the drug's effectiveness in controlling symptoms over time (Bhang et al., 2017; Lee & Je, 2018).

Formulation. However, despite the benefits of both formulation types of methylphenidate, short-acting methylphenidate is linked with poorer adherence and persistence compared to its extended-release counterpart (Ayaz et al., 2014; Bhang et al., 2017; Lee & Je, 2018). Long-acting methylphenidate offers improved symptom control with fewer doses throughout the day, making it more convenient for the patients and reducing the chances of interruption of treatment (Bhang et al., 2017; Cheung et al., 2021; Jeun et al., 2024; Lee & Je, 2018).

Switching Medication. Medication switching also played a role, with children who switched medications more than once, particularly from short-acting to long-acting formulations (Ji et al., 2018), having better adherence. However, ongoing dosage or drug adjustment can lead to treatment discontinuation due to safety issues related to side effects and inconsistent results (Ishøy et al., 2024; Titheradge et al., 2022). Finally, new medication users have higher adherence than experienced users (Brinkman, Sucharew, et al., 2018; Jeun et al., 2024; Ji et al., 2018).

Dosage and Adjustments. Higher doses can enhance the drug's effectiveness, such that patients are more inclined to adhere to their regimen (Ishizuya et al., 2021; Sobanski et al., 2014; Tuncturk et al., 2023). If the dose is not properly adjusted according to the patient's weight, this will result in poorer therapeutic efficacy, which can result in decreased motivation to remain in treatment (Sobanski et al., 2014). Tolerance breaks also decrease adherence (Barnard-Brak et al., 2023).

Side Effects. Adverse effects and side effects emerged as a major adherence barrier, with the majority of patients missing doses or stopping medication to avoid physical or mental distress (Ahmed et al., 2017; Ayaz et al., 2014; Barnard-Brak & Kudesey, 2022; Baykal et al., 2024; Brinkman, Simon, & Epstein, 2018; Druedahl & Kälvemark Sporrong, 2018; Emilsson et al., 2017, 2020; Khan & Aslani, 2020; O'Callaghan, 2014; Schein et al., 2022). Addiction and the long-term impacts of medication were another crucial issue. Due to fear of drug dependence or concerns regarding the impact of long-term medication, like stunted growth, drug abuse, or dependence (Ahmed et al., 2017; Druedahl & Kälvemark Sporrong, 2018; Khan & Aslani, 2020, 2021; Titheradge et al., 2022) some patients discontinued treatment.

Medications Benefits. Patients who had seen beneficial effects, behavioral, academic, or social interactions, were more likely to continue with medication. Positive effects, such as improvement in symptoms (Ayaz et al., 2014; Kamimura-Nishimura et al., 2022; Sobanski et al., 2014) or reduction in discomfort, has a significant contribution to the continuation of the treatment and an enhanced level of adherence. Furthermore, these therapeutic effects can generalize to other domains of living, including occupational, academic, and interpersonal relationships, as better symptom management may lead to improved work functioning, increased academic achievement, and more effective communication with others, thus creating a positive feedback loop that reinforces treatment adherence (Ahmed et al., 2017; Khan & Aslani, 2020; O'Callaghan, 2014).

Ineffectiveness of Medication. Ineffectiveness or poor response to medication is a significant concern that can reduce treatment adherence, especially in adult life (Schaefer et al., 2017). When patients feel that the treatment is not meeting their expectations or needs, they lose confidence in its efficacy and are more likely to abandon it altogether (Antony, 2016; Ayaz et al., 2014; O'Callaghan, 2014; Semerci et al., 2016; Sobanski et al., 2014).

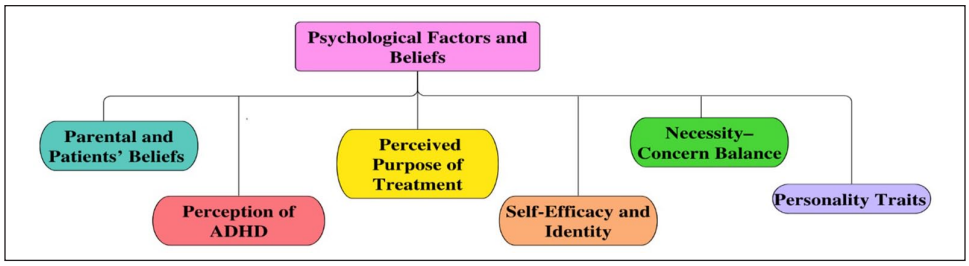


Figure 5. Subdimensions of the psychological factors and beliefs.

Co-Medication and Concomitant. Additionally, co-medication and concomitant medication like SSRIs could either positively or negatively impact adherence, depending on the combination and impact on overall treatment (Ayaz et al., 2014; Ishøy et al., 2024; Lee & Je, 2018; Mohr-Jensen et al., 2020; Titheradge et al., 2022).

In this review, these factors were the strongest predictors of adherence and ranked first among all categories. Differences between long-acting and short-acting formulations, as well as patients' perceived effectiveness or side effects, were often associated with the most substantial changes in adherence patterns.

Psychological Factors and Beliefs. In this section, six factors influencing medication adherence among patients with ADHD were uncovered. Figure 5 presents a summary of the findings in this section.

Parents and Patients' Beliefs. Patient and family beliefs and attitudes play a significant role in shaping treatment adherence (Schein et al., 2022). The perception that ADHD is a minor issue or merely a personality trait is linked to lower medication adherence, while viewing ADHD as a neurobiological disorder results in higher adherence rates (Gould & Doucette, 2018). Those who regard ADHD as a medical condition and consider stimulants as safe are more likely to start and continue medication, leading to better adherence (Charach et al., 2014). Conversely, concerns about side effects, long-term medication use, and fears of dependency may reduce adherence (Alsubaie et al., 2024; Kamimura-Nishimura et al., 2022; Khan & Aslani, 2020). Boudreau and Mah (2020) showed that positive medication opinions were associated with a higher likelihood of parents starting or continuing their child's medication.

Perceptions of ADHD. Perceptions of ADHD as a childhood or educational disorder and the belief that symptoms heal on their own can lead to medication discontinuation when individuals transition out of school (Baykal et al., 2024; Schaefer et al., 2017; Titheradge et al., 2022). Additionally, viewing treatment as a short-term intervention primarily for academic performance often results in stopping medication during weekends, school breaks, or after graduation (Ahmed et al., 2017).

Perceived Purpose of Treatment. Adolescents may discontinue treatment because they don't like the feeling of use and want to regain a sense of self, believing that the medication alters their energy levels and identity (Barnard-Brak et al., 2023; Khan & Aslani, 2020; Kosse et al., 2017; Schaefer et al., 2017). The lack of acceptance of ADHD as a legitimate condition (Khan & Aslani, 2020) or the belief that symptoms will naturally improve over time (Wagoner et al., 2020) contributed to lower adherence.

Self-Efficacy and Identity. Gould and Doucette (2018) showed that adherent students had fewer concerns about self-identity and side effects and stronger beliefs in academic and social benefits. Also, low self-efficacy and negative self-perception led to non-adherence or discontinuation (O'Callaghan, 2014).

Necessity–Concern Differential. The necessity–concern differential plays a crucial role in adherence. Individuals who perceive a greater necessity for medication compared to their concerns about potential side effects are more likely to continue treatment (Ahmed et al., 2017; Brinkman, Simon, & Epstein, 2018; Charach et al., 2014; Druedahl & Kälvemark Sporrong, 2018; Khan & Aslani, 2021; Safavi et al., 2019; Schaefer et al., 2017; Titheradge et al., 2022; Wagoner et al., 2020). Conversely, those with stronger concerns about adverse effects relative to their perceived need for medication tend to exhibit lower adherence and higher rates of intentional non-adherence (Emilsson et al., 2017, 2020).

Personality Traits. Personality traits such as antagonism and negative affectivity have been linked to lower ADHD medication adherence, with higher levels of these traits associated with increased intentional non-adherence and lower overall adherence scores (Emilsson et al., 2020).

Parents' and patients' attitudes and beliefs about ADHD and its pharmacological treatment had a profound and lasting impact on treatment continuation. A positive perception

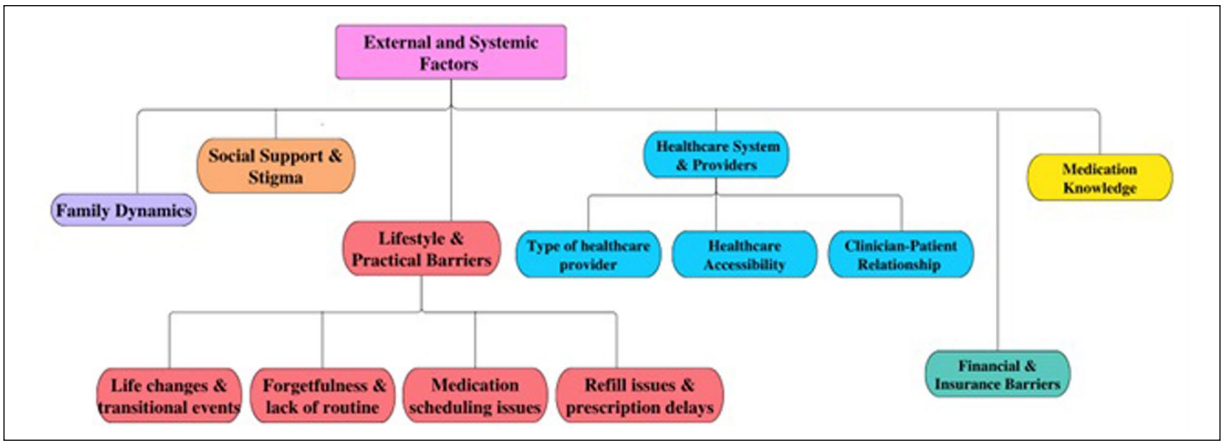


Figure 6. Subdimensions of the external and systemic factors

of the necessity of medication and alignment with personal identity strengthened adherence. These factors ranked third overall, following medication-related and systemic factors, and their importance in the chronic phase of treatment can sometimes rival that of medication-related aspects.

External and Systemic Factors. This systematic review identifies the key role of family dynamics, lifestyle concerns, and system-level barriers in affecting adherence to medication in ADHD. Figure 6 presents a summary of the findings in this section.

Family Dynamics. Family processes, including parent-child relationships, level of parental supervision, and family conflict, were all strong predictors of adherence, with positive and well-organized family environments predicting frequent medication use, whereas higher conflict, communication problems, and disagreement between parents on treatment predicted higher rates of discontinuation (Charach et al., 2014; Druedahl & Kälvemark Sporrong, 2018).

Social Support and Stigma. Having good family support systems, including community resources and support groups (Ahmed et al., 2017), played a major role in ensuring the highest possible medication adherence rates through emotional support and instrumental assistance. On the other hand, the absence of social support for parents and children led to increased stress and difficulty sustaining adherence, particularly among families that faced stigma, poverty, or limited access to healthcare services (Antony, 2016; Brinkman, Simon, & Epstein, 2018; Druedahl & Kälvemark Sporrong, 2018; Schaefer et al., 2017; Titheradge et al., 2022). Other than parenting influence, social and peer factors were implicated in adherence patterns. Teenagers with stigma due to the use of ADHD medication were more likely to drop out or miss treatment. They experienced differences from their peers and were discouraged from using medicine because of adverse attitudes from society or friends, and from their peers' pressure to give up medicine (Ahmed et al., 2017; Baykal et al., 2024; Brinkman, Simon, & Epstein, 2018; Charach et al., 2014; Druedahl & Kälvemark Sporrong, 2018; Khan & Aslani, 2020, 2021; Schaefer et al., 2017; Wagoner et al., 2020).

Lifestyle and Practical Barriers. Additionally, some adolescents intentionally adjust the timing of their medication to manage perceived social or academic stress, choosing to take medication during school days and skipping it on weekends or holidays (Gould & Doucette, 2018; Schein et al., 2022; Sleath et al., 2017; Wagoner et al., 2020). Practical barriers, such as life changes (Titheradge et al., 2022), forgetfulness (Barnard-Brak et al., 2023; Khan & Aslani, 2020; Kosse et al., 2017; Schein et al., 2022), medication scheduling issues, and lack of organized routines (Kosse et al., 2017; Schaefer et al., 2017; Schein et al., 2022), were the primary obstacles to adherence. In addition, difficulties in receiving timely refills of medication or adjusting prescriptions based on symptom changes further hinder adherence (Ayaz et al., 2014; Baykal et al., 2024; Brinkman, Simon, & Epstein, 2018; Kosse et al., 2017; O'Callaghan, 2014).

Healthcare System and Providers. Access to healthcare services and communication with providers significantly influenced adherence (Titheradge et al., 2022). Individuals who received their initial ADHD prescription from specialist psychiatric services were more likely to continue on treatment (Bhang et al., 2017; Jeun et al., 2024), whereas individuals prescribed medication within general practice or private practice were at higher risk of discontinuation (Ishøy et al., 2024). A good provider-patient relationship was associated with increased adherence levels, whereas dissatisfaction with health providers resulted in treatment drop-out (Ahmed et al., 2017; Kamimura-Nishimura et al., 2022; Khan & Aslani, 2020, 2021; O'Callaghan, 2014;

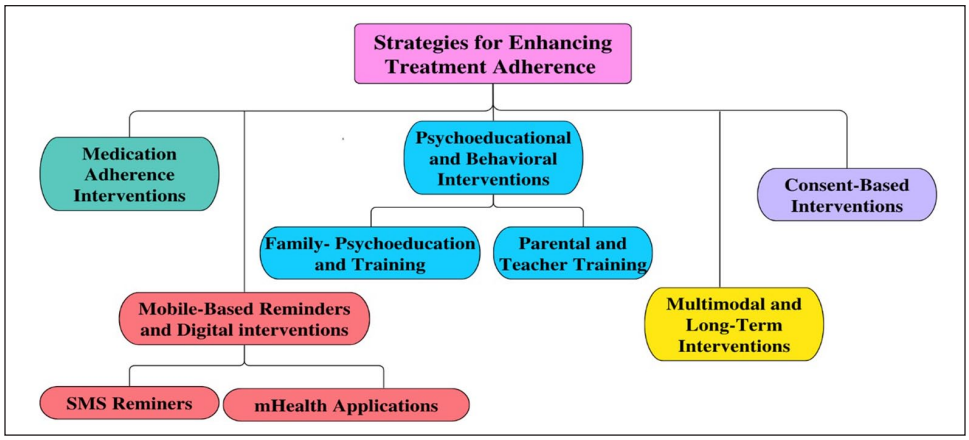


Figure 7. Subdimensions of the strategies for enhancing treatment adherence.

Sleath et al., 2017). The specialty of the treating provider also impacted adherence. Individuals treated by psychiatrists or specialists in ADHD were more adherent, while those treated by general practitioners or non-specialists exhibited higher discontinuation rates (Bhang et al., 2017). Changes in healthcare providers also disrupted continuity of care and led to erratic adherence due to differences in treatment styles, trust issues, or a lag in changing medication (Rao et al., 2021).

Financial and Insurance Barriers. Economic limitations and insurance limitations were common reasons for non-adherence. Low socioeconomic status households or uninsured households were more likely to discontinue treatment due to cost concerns. Even for the insured, out-of-pocket expenses and limited drug options available under insurance plans posed obstacles to long-term adherence (Druedahl & Kälvemark Sporrong, 2018; Jeun et al., 2024; Khan & Aslani, 2021; O'Callaghan, 2014; Safavi et al., 2019). While receiving financial aid and other benefits may also lead to reduced treatment adherence (Mohr-Jensen et al., 2020; Wagoner et al., 2020).

Medication Knowledge. Brinkman, Sucharew, et al. (2018) found that medication information and decreased decisional conflict were found to be significant predictors of short-term medication continuity. An integrated view of the different determinants of adherence to ADHD treatment is essential to achieve total understanding. Demographic and socioeconomic determinants interact with psychological determinants, drug characteristics, environmental factors, and health care system barriers to shape intricate patterns of adherence. The different determinants of adherence do not function in isolation; they are in an integrated complex network of interdependent determinants, the development of which varies across different phases of life. The determinants not only exist as isolates but also strongly interact, ultimately shaping adherence patterns.

Environmental and systemic conditions ranked second in overall importance, and their impact was especially pronounced in populations facing substantial structural barriers. External and systemic factors, particularly the quality of the patient–provider relationship and access to specialized services, often had a notable impact on adherence, although their strength varied across studies. This category typically plays an important supportive or hindering role, capable of amplifying or mitigating the effects of other factors.

Strategies for Enhancing Treatment Adherence in ADHD

Improving treatment adherence in ADHD requires a multifaceted approach that addresses key challenges such as forgetfulness, side effects, stigma, and lack of awareness. Each intervention type is supported by evidence from clinical trials and observational studies. Also, each of these strategies offers unique benefits, and their combination often yields the most substantial improvements in adherence. Figure 7 presents a summary of the findings in this section.

Medication Adherence Interventions. Pharmacologic interventions may assist with adherence through optimal treatment regimens. One study examining the addition of guanfacine extended-release (GXR) to stimulant therapy found significant improvement in adherence in pediatric and adolescent ADHD. Those treated with adjunctive GXR for 6 months demonstrated a higher Medication Possession Ratio (MPR) of 0.20 for long-acting and 0.18 for short-acting stimulants (Meyers et al., 2017).

Mobile-Based Reminders and Digital Interventions. SMS Reminders.

Short message service (text messaging) and mobile applications provide functional support by reducing forgetfulness and promoting prescription refill rates. An innovative text

messaging treatment program to prompt children to take medicines, have prescriptions refilled, and manage ADHD symptoms led to extremely significant refills of prescriptions at the appropriate time, rising from 62% in the control condition to 85% in the intervention condition (OR = 3.46, 95% CI [1.82, 6.58], $p < .05$; Fried et al., 2020). Similarly, a personalized text message program for adults reminded patients to refill prescriptions and see clinicians, resulting in increased adherence: 79% of the participants refilled prescriptions within 45 days versus 40% in the control group (OR = 5.40, 95% CI [3.14, 9.28], $p < .001$; Biederman et al., 2019). Besides short message service texts, mobile applications have proven to be effective methods. The iCON mobile app, which included medication reminders daily and symptom tracking weekly, had a higher pill count and improved ADHD symptoms in 8 weeks ($F = 4.33$, $p < .05$; $F = 4.74$, $p < .05$; Weisman et al., 2018).

mHealth Applications.

Structured digital programs that include medication monitoring, symptom recording, and education support have shown promise to enhance adherence.The FOCUS ADHD digital program includes weekly symptom ratings and side effect tracking, medication reminders, psychoeducation, treatment support network engagement, and financial rewards. Carvalho et al. (2023) found that program acceptability was 100% with the addition of financial rewards compared to 75% in the control program group. The levels of engagement, however, decreased over time for both groups.

Psychoeducational and Behavioral Interventions. Psychoeducational approaches address the cognitive and emotional factors influencing medication adherence by engaging families, caregivers, and teachers in structured education and behavioral training. Also, Psychoeducational programs aim to improve adherence by increasing awareness, addressing parental concerns, and equipping families with strategies to support medication use. These interventions frequently involve parents, teachers, or both, ensuring a more structured and supportive environment for children with ADHD.

Family-Psychoeducation and Training. Psychoeducation programs help families understand ADHD and its treatment, leading to improved adherence. The Medication Integration Protocol (MIP), a family-based psychoeducational program, significantly improved medication adherence. MIP provides psychoeducation about ADHD, its symptoms, prevalence, and impact on functioning, as well as discussing the benefits, risks, and side effects of medications. also integrates medication management activities with ongoing behavioral planning, such as creating case coordination frameworks, monitoring medication benefits and side effects, and addressing any stigma or misuse concerns.In addition to these goals, MIP is designed to help behavioral therapists take a leading role in educating clients about ADHD, promoting family-centered decisions about medication, and integrating medication management activities into behavioral treatment planning. Compared to the control group, participants in the MIP program had higher rates of psychiatric evaluation completion (100% vs. 48%) and prescription filling (57% vs. 5%; $\chi^2(1) = 12.07$, $p < .001$; Hogue et al., 2016). Similarly, a psychoeducation program based on the Theory of Planned Behavior (TPB), which included parent sessions, lectures, and peer support, resulted in lower medication discontinuation rates (13.6% vs. 46.7%) and fewer medication-free days (4.7 vs. 30.6; Bai et al., 2015). A 12-month parental psychoeducation program provided ADHD education through small group sessions, follow-up phone calls, and structured learning materials. While there was no significant difference in medication discontinuation rates, participants reported higher satisfaction and better medication adherence (Montoya et al., 2014). Another psychoeducation initiative, the G-PAM program, which provided parent and child education on medication management, led to reduced resistance to medication (SAMBA-C, $p = .05$) and improved attitudes toward medication (CAQ, $p = .026$; SAMBA-P, $p = .015$; Nagae et al., 2019).

Parental and Teacher Training. Integrating teacher training alongside parental education reinforces adherence strategies both at home and in school. A structured 6-month training program for parents and teachers significantly increased medication adherence scores, with the intervention group achieving a mean MARS score of 22.8 compared to 16.5 in the control group ($t = 5.217$, $p < .01$). The parent and teacher education program consisted of four weekly 2-hr sessions at elementary schools, covering ADHD etiology, symptoms, impairments, treatment, and follow-up. Parents were trained in medication management, behavior assessment using the SNAP-IV scale, and behavioral strategies such as positive and negative reinforcement. Teachers received similar training, with additional focus on classroom behavior management, scaffolding techniques for self-regulation, and creating emotionally supportive environments to enhance ADHD management (Zheng et al., 2020).

Multimodal and Long-Term Interventions. A comprehensive approach consisting of pharmacological, psychoeducational, and behavioral interventions has been shown to improve long-term adherence. Enns et al. (2017) assessed a 2-year multimodal intervention integrating therapy, parent support, and medication management. They found that 36% of the intervention group children achieved an MPR ≥ 0.8 compared to 23% of the control group, with a ratio of adherence rates of 1.42 (95% CI [1.03, 1.96]).

Consent-Based Interventions. Offering informed decision-making through formal consent processes can enhance adherence. Naenen-Hernani et al. (2017) found that written informed consent considerably reduced the risk of treatment discontinuation (RRR = 0.67, *SD* = 0.17) and enhanced adherence rates, where 92.5% of the participants continued treatment compared to 78% in the control group.

Note: Reported effect sizes are drawn from studies with heterogeneous designs, outcome measures, and populations; therefore, direct quantitative comparison between interventions should be interpreted with caution.

Discussion

This systematic review examined various clinical, psychological, and systemic factors influencing medication adherence in individuals with ADHD. A synthesis of the findings, based on each category's frequency and relative importance, indicates that medication-related factors hold the highest priority, showing very high importance in predicting treatment persistence—particularly formulation type, perceived effectiveness, and side effects. External and systemic factors, including the quality of the patient–provider relationship and access to specialized care, ranked second with high importance, often moderating the effects of other categories. Psychological factors and beliefs, such as patient and parental attitudes toward ADHD and its treatment, demonstrated moderate-to-high importance, strongly influencing motivation for continued adherence. Demographic and socioeconomic factors, while moderately important, generally acted as contextual or moderating influences rather than primary drivers. Finally, clinical and psychological conditions ranked lowest in relative importance, with variable effects depending on comorbidity type or symptom severity—in some cases, enhancing, and in others hindering, adherence.

Demographic factors

Demographic factors like age, gender, socioeconomic status, and education level all affect how individuals approach treatment. Younger individuals, particularly children, are generally more likely to adhere to medication regimens, likely due to greater parental oversight and involvement (Edgcomb & Zima, 2018). However, as individuals age, especially during adolescence, adherence rates tend to decrease (Ferrin et al., 2025; Gast & Mathes, 2019), driven by increasing autonomy, societal pressures, and stigma (Kamimura-Nishimura et al., 2022; Schaefer et al., 2017). Females typically observe higher adherence than their male counterparts, which might stem from differences in symptom expression or societal pressure (Ayaz et al., 2014), though hormonal changes may present unique challenges for young women (Amiri et al., 2025). In addition, individuals belonging to lower socio-economic status seem to incur unique challenges, such as financial constraints alongside limited access to health, which may impede adherence (Brinkman, Sucharew, et al., 2018; Ferrin et al., 2025).

Some of these discrepancies across studies could be explained by differing geographic regions. As Hong et al. (2013) found that East Asian pediatric patients had higher odds of non-adherence compared to Central European peers. Cultural, racial, and linguistic biases may therefore lead to differences in treatment adherence rates across groups, as cross-cultural research has indicated. This variation might be due to differing access to healthcare, cultural perceptions of treatment in mental health. For example, studies conducted in countries with more equitable healthcare systems might not observe the same socioeconomic gradients that are detected in other regions (Chan et al., 2025; Ghosh et al., 2017; Slobodin & Masalha, 2020).

Clinical and Psychological Conditions

Trials consistently demonstrate that ADHD patients who have comorbid illnesses such as Oppositional Defiant Disorder (Kamimura-Nishimura et al., 2022), tic disorders, and epilepsy (Ishøy et al., 2024; Lee & Je, 2018) are more compliant with their medication regimen, possibly because families perceive greater benefit from treatment when multiple symptoms improve (Erkan et al., 2022). But illnesses such as depression, drug use disorders, and schizophrenia have been linked with lower adherence (Ishøy et al., 2024), likely due to treatment complexity and difficulty maintaining consistent routines.

Medication-Related Factors

Medication class is a significant predictor of adherence. Long-acting stimulants, such as methylphenidate extended release, exhibit superior adherence rates compared to the short-acting types (Ayaz et al., 2014; Bhang et al., 2017), due to more stable symptom control and reduced daily dosing requirements (Cheung et al., 2021; Jeun et al., 2024). The non-stimulant, atomoxetine, has also shown higher adherence compared to short-acting stimulants, possibly due to its once-daily dosing regimen (Bhang et al., 2017; Lee & Je, 2018).

derived from studies on methylphenidate, while amphetamine-based treatments have been less frequently investigated, despite their prevalent use in adolescents and adults (Castells et al., 2018; Karlstad et al., 2016).

Also, switching medication or changing dosing may impact adherence. While some children have enhanced adherence when switching from short-acting to long-acting formulations (Ji et al., 2018), frequent medication changes—particularly when accompanied by side effects—often lead to discontinuation (Ishøy et al., 2024). Perceived efficacy of

medication and concern about side effects were consistently highlighted in studies (Coghill et al., 2017; Ferrin et al., 2025). Larger sample size studies, for example, Nayak et al. {2021) provided stronger evidence for such correlations than smaller, localized sample studies, for example, Ayaz et al. (2014). Compared to previous reviews our findings are consistent in that they highlight the multidimensional nature of adherence in ADHD.

Psychological Factors and Beliefs

Patient and family beliefs about ADHD and its treatment are central to medication adherence. Perception of ADHD as a neurobiological disorder, rather than an easy behavioral issue, leads to higher medication adherence (Gould & Doucette, 2018), while seeing it as a temporary phase or a result of poor parenting reduces adherence (Khan & Aslani, 2020; Wagoner et al., 2020). Family functioning is central to adherence to treatment. A good family climate, as characterized by high supervision and communication, promotes adherence, while high family conflict or disagreements about treatment can lead to dropout (Charach et al., 2014; Druedahl & Kälvemark Sporrong, 2018). However, underprivileged families who suffer from social stigma, financial issues, or limited healthcare access are typically disadvantaged, and this can add to the challenges of maintaining a consistent medication routine (Schaefer et al., 2017; Titheradge et al., 2022).

Beyond the immediate family, extended social support networks, such as community services and support groups, have been found to facilitate medication adherence for families who face additional adversity, such as stigma or financial hardship (Antony, 2016; Druedahl & Kälvemark Sporrong, 2018).

External and Systemic Factors

System-related healthcare factors are significant, too. Specialists, particularly psychiatrists or ADHD-trained clinicians, are better positioned to tailor treatment, adjust regimens proactively, and address patient concerns, which collectively foster trust and long-term engagement (Bhang et al., 2017; Jeun et al., 2024). This suggests that strengthening provider-patient relationships and minimizing care disruptions could be as impactful as pharmacological choices in sustaining adherence.

Aside from social and family causes, lifestyle considerations and practical constraints also play an important role in non-adherence to medication. Forgetfulness is not simply a matter of poor memory; it reflects the cognitive load of balancing school, work, and social life, particularly in adolescents and young adults with ADHD, whose executive functioning is already challenged (Khan & Aslani, 2020, 2021; Kosse et al., 2017; Schein et al., 2022). The common practice of skipping doses during weekends or holidays illustrates how adherence is shaped by beliefs about the "situational" need for medication, which can erode its long-term efficacy (Gould & Doucette, 2018; Sleath et al., 2017). This cycle pattern of drug usage can ruin its effectiveness in the long term, and therefore, the necessity of keeping a consistent drug regimen.Increasing access to healthcare services and enabling patients to easily refill medication or modify their treatment regimen is crucial in increasing adherence rates. In addition, economic limitations remain a significant challenge to adherence to ADHD treatment (Jeun et al., 2024; O'Callaghan, 2014). Overcoming these financial barriers, maybe through subsidies, improved insurance coverage, or other sources of economic support, could serve to drive significantly improved rates of adherence.

This review suggests that what influences following ADHD treatment plans usually involves many connected things, where one thing can make another stronger or weaker. For instance, low socioeconomic status (SES) not just makes it harder to get medical help, but it could also make worries about medicine safety worse, which then increases the chance of stopping treatment (Efron et al., 2020; Kalaman et al., 2023; Rieppi et al., 2002). Conversely, robust family support and better health literacy can buffer the negative impacts of side effects and stigma (Ahmed et al., 2017).

Similarly, perceived stigma can interact with adolescents' desire for autonomy, leading to intentional dose omissions—an effect that may be further intensified when side effects reduce the perceived benefit of medication (Gajria et al., 2014; Khan & Aslani, 2020). Indeed, adverse effects are among the most commonly reported proximal reasons for stopping ADHD medication; these side effects often interact with patients' concerns and social pressures (e.g., stigma), producing a stronger combined effect on non-adherence than any single factor alone (Kamimura-Nishimura et al., 2019). These patterns highlight the importance of considering the joint effects of multiple determinants rather than evaluating them in isolation.

Strategies for Enhancing Treatment Adherence

Improving medication adherence in ADHD patients needs to be addressed in a broad and multidimensional manner due to the dynamics of treatment adherence. Pharmacological adjustments, such as adding guanfacine to stimulant regimens, have been associated with improved adherence (Meyers et al., 2017). This follows the precept that pharmacological treatments, when combined with other treatments, can be effective in improving adherence in both short- and long-term scenarios (Cheng & Boileau, 2025; Treuer et al., 2013). Side effects or long-term dependence issues can limit the success of such approaches (Alsubaie et al., 2024; Khan & Aslani, 2020).

The study highlighted those Digital reminders, including text messages (Biederman et al., 2019; Fried et al., 2020) and mobile apps (Weisman et al., 2018), reminding patients about medication regimens, prescription refill times, and symptom monitoring have been encouraging in overcoming forgetfulness, which is one of the key impediments to adherence (Biederman et al., 2019; Fried et al., 2020). Such interventions leverage technology's ubiquity, rendering them convenient and easy to use, especially among younger populations who are familiar with the use of digital devices. The challenge lies, however, in maintaining long-term engagement. Carvalho et al. (2023) stated that such interventions can require regular updation or other engagement strategies to stay effective in the long term.

Psychoeducational and behavioral approaches like the Medication Integration Protocol (MIP) and Theory of Planned Behavior (TPB) interventions have been effective in having a positive influence on adherence by raising awareness of ADHD and treatment knowledge gaps, building routine, and equipping caregivers to manage side effects (Bai et al., 2015; Hogue et al., 2016). Zheng et al. (2020), provide a more comprehensive solution, targeting both home and school environments, which are crucial to children with ADHD. The success of these programs suggests that involving the most significant stakeholders in a child's life, both at home and at school, yields a more conducive environment for medication adherence. Psychoeducational interventions might not be effective for all families, based on socio-economic status, education level, and prior knowledge regarding ADHD (Druedahl & Kälvemark Sporrong, 2018). Plus, psychoeducational interventions for ADHD are more commonly applied in younger children, often effectively addressing the issues and reducing, or even eliminating, the perceived need for medication (Feldman et al., 2018; Rodrigo Jiménez et al., 2022; Syed et al., 2024).

A multimodal approach that combines pharmacological treatment with psychoeducational and behavioral treatments is very effective in promoting long-term adherence. The Enns et al. (2017) A study illustrated that children under a 2-year multimodal treatment had improved adherence rates compared to those under usual care. The findings of the MTA indicate that while medication significantly reduces ADHD symptoms in the short term, it alone is insufficient to restore functioning to the level of healthy peers, particularly among children and adolescents, and its effectiveness tends to diminish over time; this underscores the necessity of combined approaches—integrating structured behavioral interventions and environmental supports, as implemented in the multimodal MTA protocol—to sustain and enhance treatment outcomes in the long term (Döpfner et al., 2020; Li, 2024; Sibley et al., 2022). Also, ensuring informed decision-making through consent-based models can enhance adherence. Naenen-Hernani et al. (2017) highlighted that requiring written informed consent significantly reduced the risk of treatment discontinuation.

This strategy fosters a sense of ownership and commitment to treatment, which can be particularly important for adolescents and young adults, who may otherwise feel disconnected from their treatment plan. Several findings in the included studies are supported by existing literature. For instance, the effectiveness of digital tools in ADHD treatment adherence aligns with prior evidence suggesting that mHealth tools significantly reduce medication gaps in chronic conditions (Hamine et al., 2015). However, unlike some other digital health research that found long-term engagement challenging (Vitija et al., 2022), this review identified programs like FOCUS ADHD (Carvalho et al., 2023) maintaining relatively high engagement rates, especially when reinforced with financial incentives.

There have also been discrepancies in the effectiveness of adherence improvement interventions. For example, Montoya et al. (2014) reported no significant difference in discontinuation rates despite psychoeducation, diverging from Hogue et al. (2016) and Bai et al. (2015), who observed strong positive effects. This may be explained by differences in intervention intensity, session length, and population demographics. Montoya's sample was drawn from a European population, potentially reflecting cultural norms of medication skepticism, while Bai''s intervention in China involved extensive parental involvement and peer support, possibly enhancing perceived medication value (Sethia, 2024). Clinically, what these findings mean is that adjustment of adherence methods to patient characteristics—age, family engagement, digital literacy, and cultural setting—is at the forefront. Physicians must integrate digital methods for short-term benefits of adherence, but pair them with the use of chronic psychoeducational or behavioral interventions to create long-term success.

Strengths and Limitations

Our study has several strengths. First, the study pools a wide sample of empirical material—52 varied study designs, ranging from cross-sectional and longitudinal studies to experiments and qualitative studies. Heterogeneity in methodology adds strength to the validity of the conclusions as well as encourages a multifaceted understanding of treatment adherence of ADHD across age ranges and intervention methods. Second, the review presents a synthesized method that reflects the sophistication of adherence behavior by grouping influential factors into five broad categories: demographic, socioeconomic, clinical, psychological, medication-related, cognitive/attitude, and systemic/environmental. Further, analysis goes beyond simply listing the factors by offering considered descriptions of the ways these domains cut across each other and shift throughout the lifecycle, from childhood to older ages. Third, the integration of both psychosocial and pharmacological interventions for the encouragement of adherence into the review is also an essential contribution.

The majority of reviews of these fields cover them separately, whereas this study compares them simultaneously, thereby presenting a fuller picture of effective interventions. In particular, emphasizing psychoeducational, digital, and consent-based interventions captures an up-to-date, patient-centered approach to treatment adherence that fits contemporary trends in the provision of mental healthcare. By and large, this review stands out for its clarity, comprehensiveness, and synthesis of a broad evidence base into a compelling narrative. It not only nominates the most suitable adherence-related variables but also emphasizes the necessity of tailoring interventions to the individual's developmental, psychological, and social situation.

Our study, nevertheless, has several limitations. First, despite the inclusion of 52 studies with diverse methodologies and populations, the heterogeneity in study designs, outcome measures, and adherence definitions posed challenges for direct comparisons and meta-analytic synthesis. Second, although this review aimed to encompass a broad age range, from children to adults, there remains a relative paucity of high-quality studies focusing on adult populations. Third, while the review included both pharmacological and non-pharmacological interventions, the evidence base for non-pharmacological strategies remains limited and heterogeneous. Previous meta-analyses have similarly reported mixed findings and methodological shortcomings in studies evaluating behavioral and psychosocial interventions (Chaulagain et al., 2023). Fourth, reliance on self-report measures in a majority of studies presents concerns regarding social desirability and recall bias. sixth, a narrow range of medication comparisons is available in the existing literature. We found no studies that directly compare methylphenidate and amphetamine formulations, which limits our ability to assess potential differences in adherence and tolerability between these two commonly used stimulant classes (Hodgkins et al., 2012; Setyawan et al., 2013). Although both types are widely recognized in clinical practice, systematic evidence of comparative adherence patterns appears limited. Likewise, comparing stimulant and non-stimulant medicines (like atomoxetine) was hard and produced inconsistent results regarding adherence and persistence (Ferrin et al., 2025; Greven et al., 2017; Radonjić et al., 2023). Because of the small number of studies and the heterogeneity in methodology, it's hard to reach strong conclusions. So, these missing pieces in the data, not how we searched, are why we can't fully judge group-specific patterns in adherence to the plan.

Clinical Implications

This systematic review highlights the reality that adherence to medication in ADHD remains a significant clinical problem, particularly in adolescence. Despite documented evidence for the efficacy of pharmacologic and psychosocial interventions, adherence continues to negatively affect maximum benefits. Our study shows that an enormous array of variables influence adherence, from patient age and psychological comorbidity to attitudes about medication and family, as well as system involvement such as accessibility to healthcare and provider interaction. Considering the multifaceted nature of adherence, improved outcomes require an integrated and individualized approach. So, this systematic review highlights several practical strategies that clinicians can adopt to enhance treatment adherence in individuals with ADHD: Educating patients and their families about ADHD and its treatments can significantly improve adherence. The priority for psychoeducation should be placed by clinicians for both caregivers and patients, and consider adding behavioral and technological assistance to routine treatment (Khan & Hasan, 2024; Syed et al., 2024). The use of smartphone applications and digital interventions has shown promise in supporting ADHD management. Recent research suggests that app-based therapies can serve as supplemental tools, offering reminders and educational content to assist with adherence (Ha et al., 2022; Hosseinnia et al., 2025; Vitija et al., 2022). Moreover, long-acting formulations and co-located behavioral health services can enable ongoing treatment engagement. Involving patients and their families in treatment decisions fosters a collaborative environment, leading to increased satisfaction and adherence (Brinkman et al., 2013). Providers must not only monitor symptom reduction but are also responsible for actively monitoring patients' and families' beliefs regarding medication benefits and side effect worries. Family participation must be encouraged, especially in children, and medication regimens streamlined wherever possible to enable usability. Also, all medication adherence interventions need to be age-tailored, and particular attention should be paid to the unique difficulties facing adolescents. By implementing these evidence-based strategies, clinicians can enhance treatment adherence among individuals with ADHD, leading to improved clinical outcomes and quality of life.

Future Research Directions

Certainly, based on the current literature and identified gaps, the following future research directions are recommended to enhance treatment adherence in individuals with ADHD. Further investigation into behavioral therapies, psychoeducation, and peer support models is essential. These interventions may offer alternative or complementary strategies to medication, particularly for individuals who experience side effects or prefer non-pharmacological approaches (Kamimura-Nishimura et al., 2019). This is while research into the effectiveness of digital tools, such as mobile applications and wearable devices, in monitoring and promoting adherence is warranted. For instance, the

AIMI system utilizes smartphone sensors and machine learning to predict medication adherence, demonstrating promising results in preliminary studies (Mamun et al., 2025). There is a need for long-term studies assessing the sustained effects of ADHD medications on quality of life and functional outcomes. While short-term benefits are documented, the long-term impact remains under-researched. Also, Research is needed to develop strategies that support individuals with ADHD during critical transition periods, such as moving from adolescence to adulthood. Effective transition care can prevent lapses in treatment adherence and ensure continuity of care. By pursuing these research avenues, the field can move toward more effective and individualized strategies to improve treatment adherence in ADHD, ultimately leading to better patient outcomes.

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ORCID iDs

Mobina Karimi https://orcid.org/0009-0003-5345-7450
Sogand Ghasemzadeh https://orcid.org/0000-0003-0897-1568
Faezeh Shabanali Fami https://orcid.org/0000-0003-4959-5271

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This study is a systematic review and did not involve any direct human or animal participation. Therefore, ethical approval was not required.

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Supplemental Material

All articles incorporated in this systematic review are accessible online. The supplementary files contain detailed search strategies, quality assessment methods, and additional data extracted from the reviewed studies, most of which are described in the methodology section.

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Author Biographies

Mobina Karimi is a graduate in Psychology and Education of Exceptional Children from the University of Tehran. Her research focuses on treatment and treatment adherence in children and adolescents with neurodevelopmental disorders, particularly Attention-Deficit/Hyperactivity Disorder (ADHD). Her interests center on expanding knowledge and deepening understanding of the psychological and behavioral aspects of children and adolescents, especially in relation to their development, learning, and quality of life.

Sogand Ghasemzadeh, PhD, is an Associate Professor in the Department of Psychology and Education of Exceptional Children at the University of Tehran, Iran. Her research focuses on the assessment and transdiagnostic treatment of behavioral and emotional neurodevelopmental disorders, particularly ADHD. Dr. Ghasemzadeh's interdisciplinary studies in digital health innovation have been recognized with several research awards. She actively contributes to advancing mental health initiatives and developing integrative, technology-assisted therapeutic strategies for children.

Faezeh Shabanali Fami, MSc, is a graduate researcher in the Faculty of Psychology and Educational Sciences at the University of Tehran, Iran. Her interdisciplinary research integrates digitechnology to address challenges in neurodevelopmental disorders, especially specific learning disorders and ADHD. She is interested in innovative technology-based interventions to enhance cognitive and behavioral functioning, academic performance, and emotional well-being in children.